

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

SOME BASIC CONCEPT OF CHEMISTRY

LECTURE-5

CHEMISTRY

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Baaz
Batch
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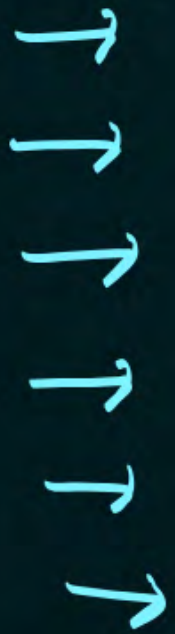
Topics To Covered

- 1** Relative atomic mass
- 2** Average atomic mass
- 3** Molecular mass and formula mass
- 4** Mole

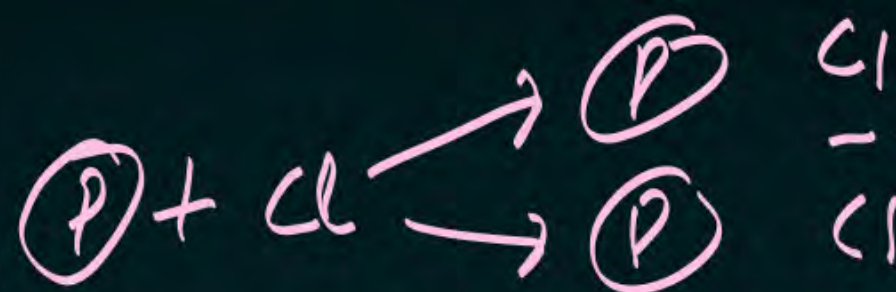
REVISION $\Rightarrow$

- 3) Law of multiple proportion 4) Gay Lussacs law 5) Avagadro Law.

Dalton's Atomic Theory



QUESTION



Phosphorus and chlorine form two compounds. The first compound contains 22.54% by mass of phosphorus and 77.46% by mass of chlorine. In the second compound the percentages are 14.88% for phosphorus and 85.12% for chlorine. Show that these are consistent with the law of multiple proportions.

Sol: ^{Case 1} Mass of Phosphorus = 22.54 g.
" " Chlorine = 77.46 g.
22.54 g of 'P' combines with = 77.46 g of Cl.

1 g of P

 " " = $\frac{77.46}{22.54}$ g of Cl.
= 3.43 g of Cl

Case 2 Mass of P = 14.88 g
" " Cl = 85.12 g
14.88 g of 'P' combine with = 85.12 g of Cl.

1 g " " "

 " " = $\frac{85.12}{14.88}$ g of Cl
= 5.72 g of Cl.

Ratio of mass of Cl = $\frac{3.43}{5.72} \approx \left(\frac{3}{5}\right)$, hence follow.

$$= 0.59 = 0.5 = \frac{1}{2}$$

Relative Atomic mass $\rightarrow$

It is not the exact atomic mass of an atom. It is defined as.

"how many times an atom of an element is heavier than $\frac{1}{12^{\text{th}}}$ of an atom of ^{12}C ".

Mathematically,

$$\text{Relative Atomic Mass} = \frac{\text{Mass of an atom}}{\frac{1}{12^{\text{th}}} \text{ of mass of an atom of } ^{12}\text{C}}$$

=

• Its scale is known as atomic mass unit scale (amu).

• $1 \text{ amu} = \frac{1}{12^{\text{th}}}$ of mass of an atom of ^{12}C .

$$= 1.66 \times 10^{-24} \text{ g or } 1.66 \times 10^{-27} \text{ kg.}$$

• Standard used for calculating relative atomic mass is ^{12}C . because relative atomic mass values calculated from this mostly or approximately give whole number.

• Now a days, "amu" is sometimes replaced with "u" (unified mass).

For example let us calculate the relative atomic mass of hydrogen atom have 1.67×10^{-24} g mass.

$$\therefore \text{Relative atomic mass} = \frac{\text{Mass of an atom of hydrogen}}{\frac{1}{12}^{\text{th}} \text{ of mass of } {}^R\text{C atom. (1 amu)}}$$

$$= \frac{1.67 \times 10^{-24} \text{ g}}{1.66 \times 10^{-24} \text{ g}}$$

$$= 1.008 \text{ amu}$$

$$= 1 \text{ amu.}$$

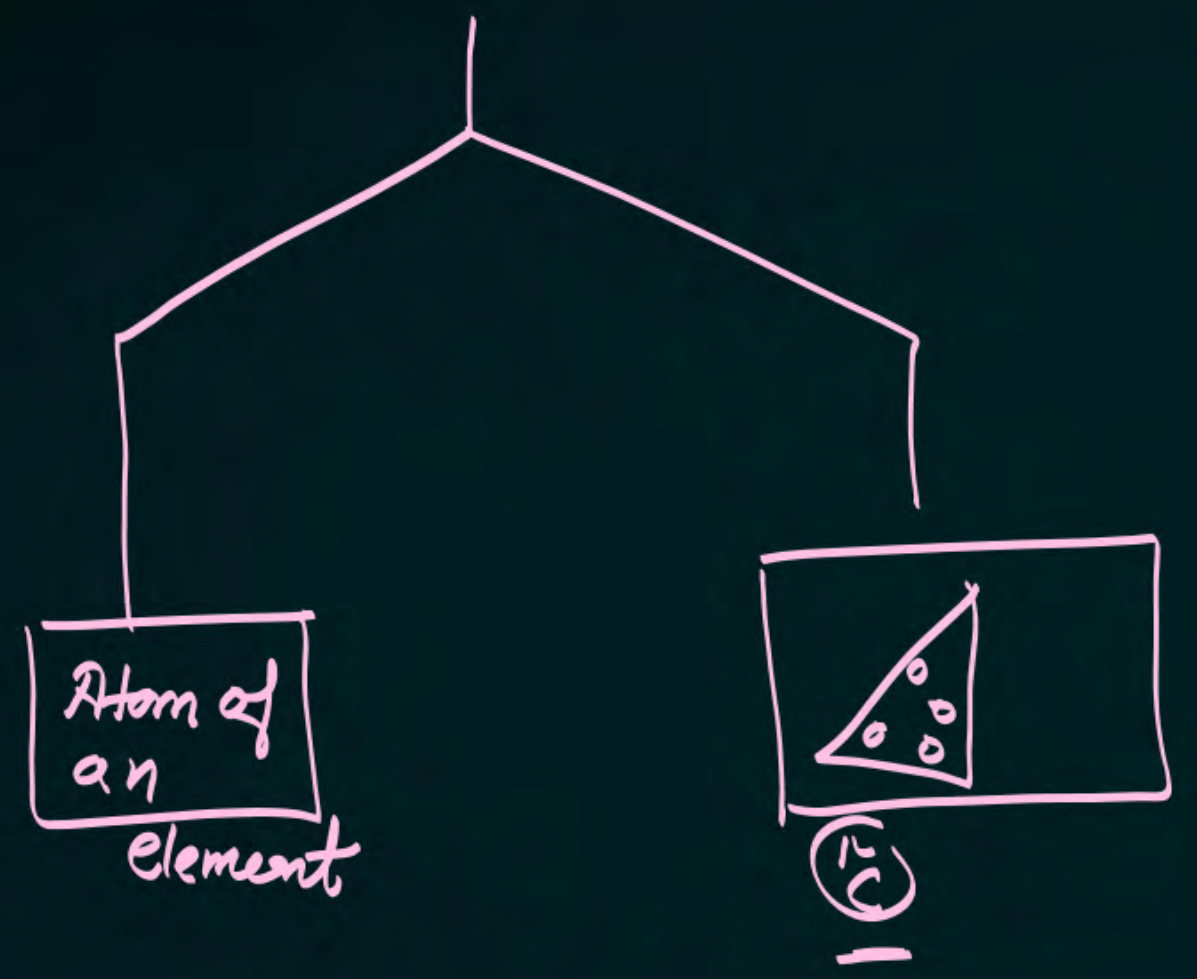
Chizzi

$^{12}_6\text{C}$

$^{12}_6\text{C}$
 $\rightarrow$
 $^{13}_6\text{C}$
 $\rightarrow$
 $^{14}_6\text{C}$



$\frac{1}{12}$ th of $^{12}_6\text{C}$



Average atomic mass



Few elements that exist in more than one isotopes, their relative atomic mass is calculated by taking average of different isotopes. Average atomic mass is calculated with help of relative atomic mass & Relative abundance of particular isotope.

*

Relative abundance:→

It is the percentage in which a particular isotope exist in nature.

Average atomic mass

Lalnu Sawal.



For example, carbon occurs as three isotopes with relative abundances and masses as:

Isotope		Relative abundance (%)	Atomic mass (amu)
(1)	^{12}C	$(98.892 \times$	$12) +$
(2)	^{13}C	$1.108 \times$	$13.00335) +$
(3)	^{14}C	$(2 \times 10^{-10}$	$14.00317)$

Therefore, average atomic mass of carbon $= \frac{(98.89 \times 12) + (1.10 \times 13) + (2 \times 10^{-10} \times 14)}{100}$
 $= 12.01 \text{ amu.}$

Calculation of average atomic mass

pyq

Calculate the atomic mass of naturally occurring argon from the following data:
Isotope

Isotope	Isotopic mass (g mol ⁻¹)	Abundance
³⁶ Ar	(35.96755 ×	0.337%) +
³⁸ Ar	(37.96272 ×	0.063%) +
⁴⁰ Ar	(39.9624 ×	99.600%)

$$\begin{aligned}
 \text{Average atom mass} &= \frac{(0.337 \times 35.96) + (0.063 \times 37.96) + (99.60 \times 39.96)}{100} \\
 &= 39.9 \text{ amu} \\
 &\approx 40 \text{ amu}
 \end{aligned}$$

Home work



Calculate the atomic mass (average) of chlorine using the following data:

%	Natural Abundance	Molar Mass
^{35}Cl	75.77	34.9689
^{37}Cl	24.23	36.9659

Ans.
35.5

Element Name	Atomic Number	Atomic Mass	Element Name	Atomic Number	Atomic Mass
* Hydrogen	1	<u>1</u>	* Sodium	11	23
Helium	2	4	* Magnesium	12	24
Lithium	3	7	Aluminium	13	27
Beryllium	4	9	Silicon	14	28
Boron	5	11	* Phosphorus	15	31
* Carbon	6	12	* Sulphur	16	32
* Nitrogen	7	14	* Chlorine	17	35.5
* Oxygen	8	16	* Argon	18	40
Fluorine	9	19	* Potassium	19	39
Neon	10	20	* Calcium	20	40

Molecular mass

Molecular mass is the sum of atomic masses of elements present in a molecule.

Example:

$$\begin{aligned} H_2O &= (2 \times 1) + 16 \\ &= 18 \text{ amu.} \end{aligned}$$

$$(a) \quad CH_4 = 12 + 4(1) = 16 \text{ amu}$$

$$(b) \quad C_6H_{12}O_6 = (6 \times 12) + (12 \times 1) + (6 \times 16) = 72 + 12 + 96 = 180 \text{ amu.}$$

$$(c) \quad NH_3 = 14 + 3(1) = 17 \text{ amu.}$$

$$(d) \quad CO_2 = 12 + (2 \times 16) = 12 + 32 = 44 \text{ amu.}$$

Gram molecular mass or gram atomic mass

~~Gram molecular mass~~

When atomic mass of any molecule or an atom is expressed in terms of gram instead of amu, then it is known as Gramic molecular mass or Gram atomic mass.

Example.

Gram atomic mass of O = 16g

Gram molecular mass of CO_2 = 44g

Gram molecular mass of H_2O = 18g.

Formula mass

- Ionic compounds such as NaCl , KNO_3 , Na_2CO_3 etc. do not consist of molecules i.e., single entities but exist as ions closely packed together in three dimensional space.

• For these compounds, formula mass is calculated & is calculated as

e.g Formula mass of $\text{NaCl} = 23 + 35.5$
 $= 58.5 \text{ amu or u}$

$\text{KNO}_3 =$

$\text{Na}_2\text{CO}_3 =$

$\text{KCl} =$

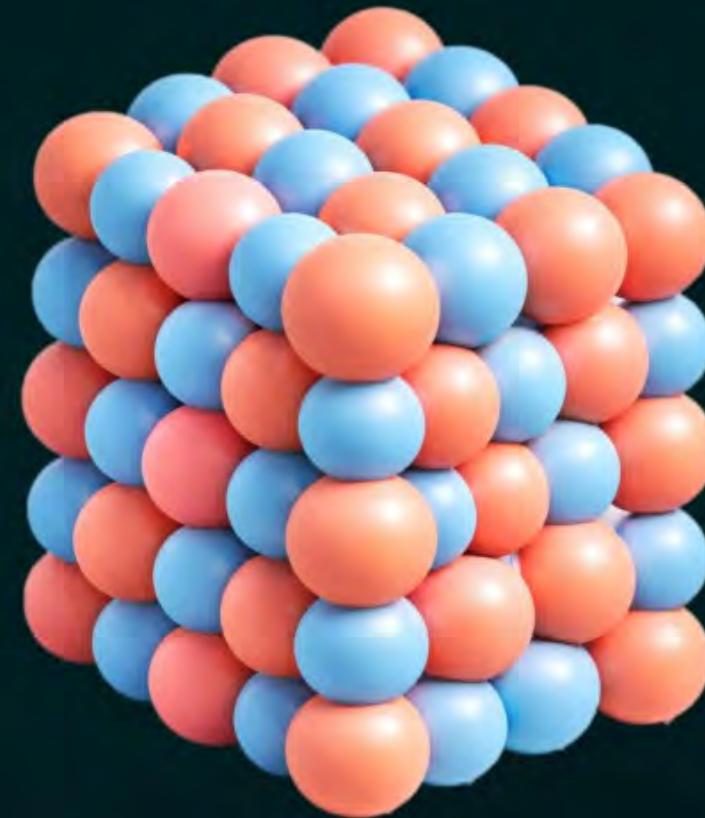


Figure: Packing of Na^+ and Cl^- ions in sodium chloride

Limitation of Dalton's Atomic Theory

The main drawbacks of Dalton's atomic theory are:

- (i) It could explain the laws of chemical combination by mass but failed to explain the law of gaseous volumes. *(Gay Lussac).*
- (ii) It could not explain why atoms of different elements have different masses, sizes, valencies, etc.
- (iii) It could not explain how and why atoms of different elements combine with each other to form compound atoms or molecules.
- (iv) It failed to explain the nature of forces that bind together atoms in a molecule
- v) *It does not explain anything about isotope & isobars.*

QUESTION



The modern atomic weight scale is based on:

1 C^{12}

2 O^{16}

3 H^1

4 C^{13}

QUESTION



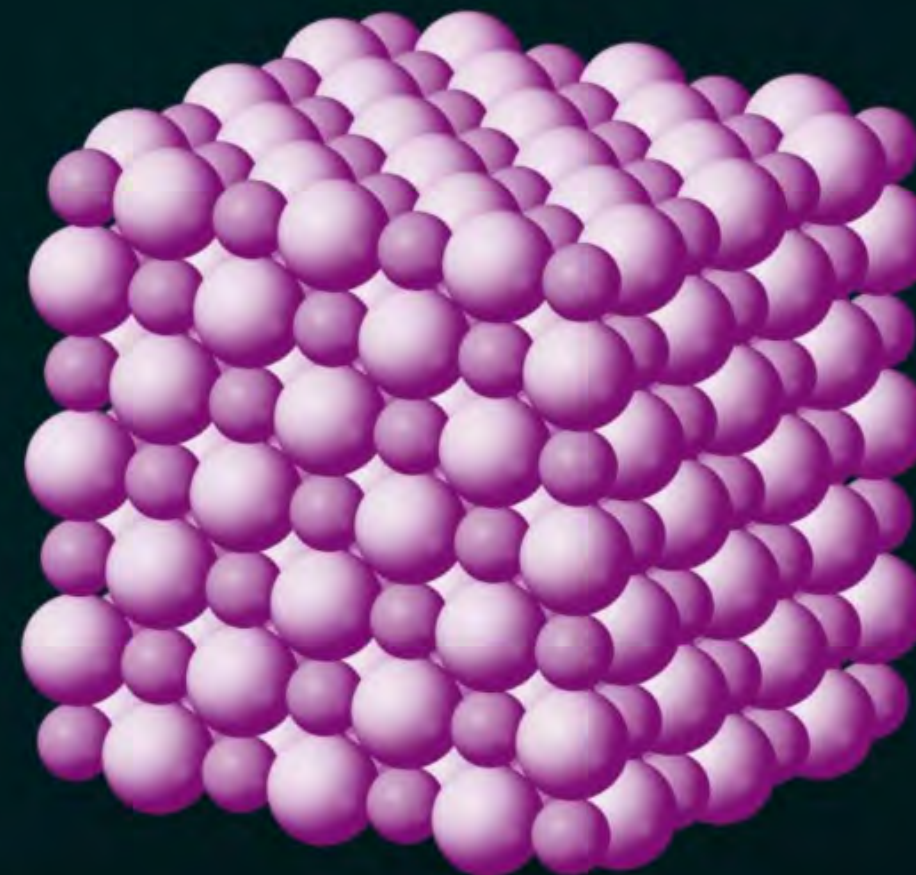
Packing of Na^+ and Cl^- ions in sodium chloride is depicted by the given figure. Choose the correct option regarding formula mass of sodium chloride.

1 In the solid state, sodium chloride does not exist as a single entity.

2 Formula mass of NaCl is 68.0 u .

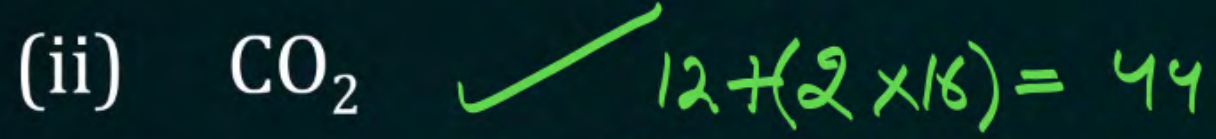
3 Formula mass of NaCl is the sum of atomic masses of Na and Cl .

4 Both (1) and (3).



QUESTION

Calculate the ~~molar~~ molar mass of the following



1 18, 44, 16

2 18, 22, 8

3 ~~9, 44, 16~~

4 ~~44, 18, 16~~

QUESTION

Which of the following statements indicates that law of multiple proportion is being followed?

- 1 Sample of carbon dioxide taken from any source will always have carbon and oxygen in the ratio 1 : 2.
- 2 Carbon forms two oxides namely CO_2 and CO , where masses of oxygen which combine with fixed mass of carbon are in the simple ratio 2 : 1.
- 3 When magnesium burns in oxygen, the amount of magnesium taken for the reaction is equal to the amount of magnesium in magnesium oxide formed.
- 4 At constant temperature and pressure, 200 mL of hydrogen will combine with 100 mL oxygen to produce 200 mL of water vapour.

Summary



- ① Soliha
- ② Naddem hone
- ③ Maseeb Sheikh

Ar 40

Ca 40

Thank
You